

Demonstrating VibraForge: An Open-source Vibrotactile Prototyping Toolkit with Scalable Modular Design

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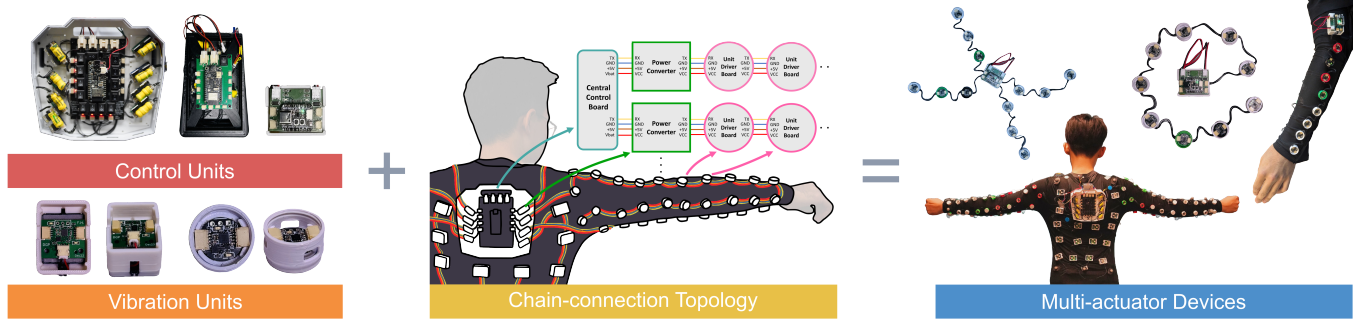


Figure 1: VibraForge is an open-source vibrotactile toolkit that comprises various types of control units and vibration units. Units are connected via chain connections to build multi-actuator systems, such as vibration rings, armbands, and full-body suits.

ABSTRACT

We demonstrate VibraForge, an open-source vibrotactile prototyping toolkit¹ that supports fine-grained control of up to 120 vibrotactile actuators. Our solution features modular and scalable design principles, self-contained vibration units, chain-connection topology, and custom communication protocol. Additionally, we offer a GUI editor for intuitive multi-actuator pattern authoring.

¹<https://github.com/huangbj16/VibraForge>

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UIST Adjunct '24, October 13–16, 2024, Pittsburgh, PA, USA

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ACM ISBN 979-8-4007-0718-6/24/10

<https://doi.org/10.1145/3672539.3686764>

This toolkit significantly lowers the barriers to haptic design and expands the design space for multi-actuator applications.

In our demonstration, participants will have the chance to experience the full process of designing vibrotactile systems from scratch using VibraForge. This includes assembling vibrotactile wearable devices, mounting them on clothes, designing vibration patterns, and real-time evaluation. Our goal is to showcase the transformative potential of multi-actuator systems and lower the barriers to haptic design for researchers and designers.

CCS CONCEPTS

• **Human-centered computing** → **Haptic devices; User interface toolkits.**

KEYWORDS

Haptics, Vibrotactile, Toolkits, Wearable Devices

ACM Reference Format:

Bingjian Huang, Hanfeng Cai, Siyi Ren, Yeqi Sang, Qilong Cheng, Paul H. Dietz, and Daniel Wigdor. 2024. Demonstrating VibraForge: An Open-source Vibrotactile Prototyping Toolkit with Scalable Modular Design. In *The 37th Annual ACM Symposium on User Interface Software and Technology (UIST Adjunct '24)*, October 13–16, 2024, Pittsburgh, PA, USA. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3672539.3686764>

1 INTRODUCTION

Vibrotactile feedback [3], which produces tactile sensations via vibrations on the skin, have gained much popularity because they are compact, affordable and accessible. Vibrotactile systems are widely adopted in various domains, such as virtual reality [8, 15], robotics [1, 11], medical technology [6, 7], and accessibility [9, 16].

To support rapid design of vibrotactile systems, researchers have developed several prototyping toolkits [4, 10, 12, 13]. In these toolkits, a central microcontroller (e.g., Arduino, RaspberryPi, ESP32, etc) is directly connected to multiple vibrotactile actuators. The central microcontroller receives inputs via audio signals or haptics waveform editor, and directly control the vibration waveform of each actuator. For example, VITAKI [10] is a vibrotactile controller that controls eight eccentric rotating mass (ERM) actuators with PWM drivers, designed for virtual reality and video games usage. Syntacts [12] is an audio-controlled haptics platform that allows designers to design vibration patterns on a GUI and send them to actuators. VHP [4] provides a custom PCB board that uses 12 PWM channels to drive actuators and an additional current sensing module to detect running performance. SleeveIO [13] provides a modular and reconfigurable platform for multimodal tactile design. These toolkits support a wide range of actuators and are designed to be user-friendly, making them suitable for both professional prototyping and educational purposes.

In recent years, the application of tactile feedback in increasingly complex scenarios has necessitated the use of a greater number of actuators. For instance, in the realm of virtual reality, comprehensive full-body tactile feedback is essential to attain an immersive experience akin to that depicted in "Ready Player One." Similarly, in the field of robot teleoperation, substantial arrays of actuators are crucial for providing the operator with detailed feedback on the robot's performance and environmental interactions, such as nearby obstacles [5, 17]. Moreover, in medical contexts, actuator grids are employed to mitigate tactile deficiencies associated with prosthetic devices [6].

As the demand for multi-actuator applications increases, existing toolkits often fail to scale effectively, resulting in higher costs, complex wiring, and reduced vibration detail. This issue primarily arises because actuators are driven directly by the central microcontroller, which is limited by the number of available IO pins and PWM signal generators on chip. Additionally, the inclusion of more actuators invariably impacts system latency and reliability. While some researchers have turned to commercial solutions like bHaptics [2], these products typically offer limited control and lack flexibility in configuration.

In our demonstration, we showcase VibraForge, an open-source vibrotactile prototyping toolkit with scalable modular design. Each vibration unit is a self-contained module with an actuator driven by its own microcontroller. Various types of vibration units are

designed, including an x-axis LRA for body extremities and a z-axis VCA for the body trunk. Vibration units are connected through chain-connection topology, where arbitrary number of actuators can be grouped as a chain and connected to a central control unit using only one IO pin. Data communication is done through custom UART protocol with high-bandwidth and low-latency. To better support vibration pattern authoring, we also designed a GUI editor that supports intuitive control of individual actuator, as well as multi-actuator pattern generation. Together, these features unveil the design space of multi-actuator applications which remain largely unexplored. Our goal is to showcase the transformative potential of multi-actuator systems and lower the barriers to haptic design for researchers and designers.

2 VIBRAFORGE DESIGN

We provide an overview of the design of the VibraForge toolkit, including vibration unit, control unit, chain-connection topology, and GUI editor.

2.1 Vibration Unit

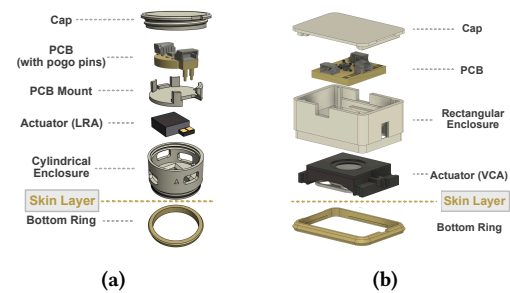


Figure 2: Exploded views of (a) the LRA vibration unit and (b) the VCA vibration unit.

Each vibration unit in the VibraForge toolkit is a self-contained module that includes a vibrotactile actuator, a custom-designed PCB, and a 3D-printed enclosure (Figure 2).

The 3D-printed enclosure was designed to enclose the vibrotactile actuator and PCB board stably and provide garment attaching method. Similar to the size of a coin, the enclosure is compact and lightweight, enabling comfortable wearing experience. The enclosure consisted of four parts: cap, PCB mount, cylindrical enclosure, and bottom ring.

Each vibration unit incorporates a dedicated driver PCB to receive UART commands from the central board, enabling precise control over vibration parameters. This driver board consists of a PIC16F18313 Microcontroller Unit (MCU), a DRV8837 H-bridge motor driver, two JST headers for power and signal connections, and two pogo pins. When the MCU received a "start" signal, it sent waveform to the H-bridge to directly modulate the actuator. We utilized sPWM waveform generator inside the MCU to provide fine-grained control over intensity, frequency and waveform.

The units are engineered to be compact, lightweight, and comfortable for wearing experience. The key to their design is versatility and ease of integration, allowing for the support of different actuator types—X-axis LRA, Z-axis VCA, etc. This helps expand their

suitability to various body positions and their ability to deliver distinct tactile perceptions required by different applications.

2.2 Control Unit

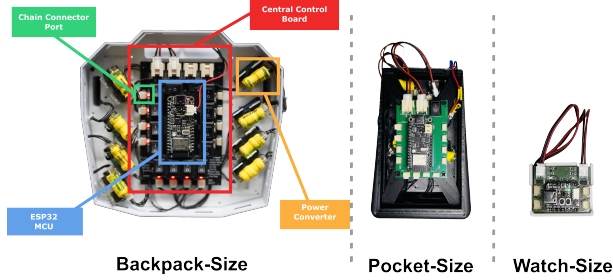


Figure 3: Three sizes of control units that support 12, 8 and 4 chains respectively. The close-up view of the backpack unit describes the positions of the central control board, chain connection ports, power converters, and ESP32 MCU.

The control unit (Figure 3) acts as the central command hub for the VibraForge toolkit. It is capable of managing vibration units across multiple chains, with each chain handling up to 30 individual units. The central unit itself is powered by two 3.7V LiPo batteries and uses an ESP32 MCU² for processing Bluetooth signals into serial commands. This ensures that the whole system can run wirelessly, which is crucial for wearable design. The control unit design also includes several versions to accommodate different scales of projects, from full-body suits to smaller, localized applications like arm bands. Each version is tailored in terms of the number of chains it supports, the model of the ESP32 board used, and the battery capacity, ensuring flexibility across various usage scenarios.

2.3 Chain-connection Topology

We implemented a chain-connected topology (Figure 4) to reduce wiring complexity and enable independent control of each vibration unit. In this configuration, vibration units are connected like a chain and only the first vibration unit directly interfaces with the central unit. Control commands, issued via UART, propagate through the chain to the target unit based on its unique address. This approach allows a single GPIO pin on the MCU to control up to 30 units, thus overcoming the limitations of GPIO availability in previous vibrotactile systems. By integrating a custom UART protocol, our design achieves low-latency, precise control over the vibration parameters of individual units.

2.4 GUI Editor

The GUI Editor (Figure 5) is designed to democratize the creation of multi-actuator vibrotactile patterns. It supports integration with existing software tools like Syntacts [12] and VIREO [14], allowing users to import and utilize pre-existing waveform designs. The GUI features a straightforward layout with a waveform template library, a central canvas for arranging actuators, and a timeline for synchronizing vibrations across multiple units. This setup enables users

²<https://learn.adafruit.com/adafruit-esp32-feather-v2>

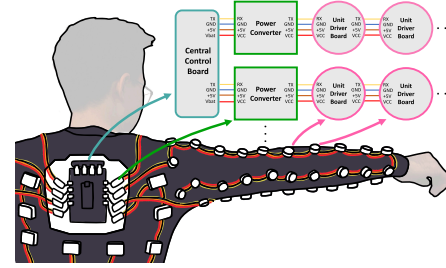


Figure 4: The multi-chain-connection topology of NebulaSuit, with the detailed connection diagrams for each chain.

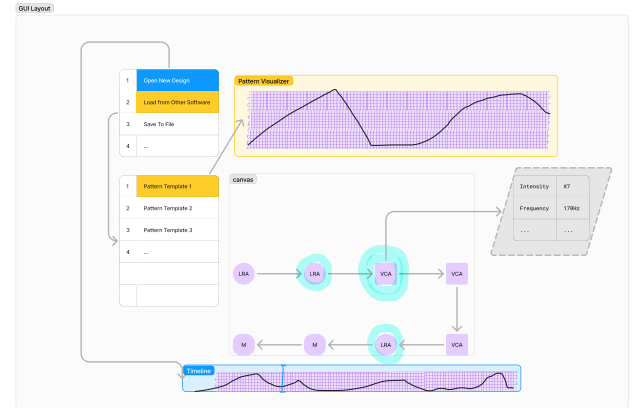


Figure 5: Example layout of the GUI editor (in progress)

to design complex haptic interactions without prior programming knowledge.

3 DEMONSTRATION DESIGN

An interactive environment (Figure 6) will be setup to demonstrate the capabilities of the VibraForge toolkit and engage potential users in its practical applications. This space will be strategically designed to not only showcase the toolkit's ease of use and versatility but also to gather user feedback on its robustness and utility.

3.1 Demonstration Space Layout

The demonstration area will be organized with several key components to facilitate interaction and learning:

- **Project poster:** A large, visually appealing poster will be prominently displayed to provide a quick overview of the VibraForge toolkit, highlighting its key features and benefits. This poster will serve as an initial engagement point for attendees.
- **Introduction video:** A monitor will play a looping video that introduces the VibraForge toolkit, illustrating its development, intended use cases, and potential impact on vibrotactile technology. The video will help in setting the context and enhancing understanding among the audience.

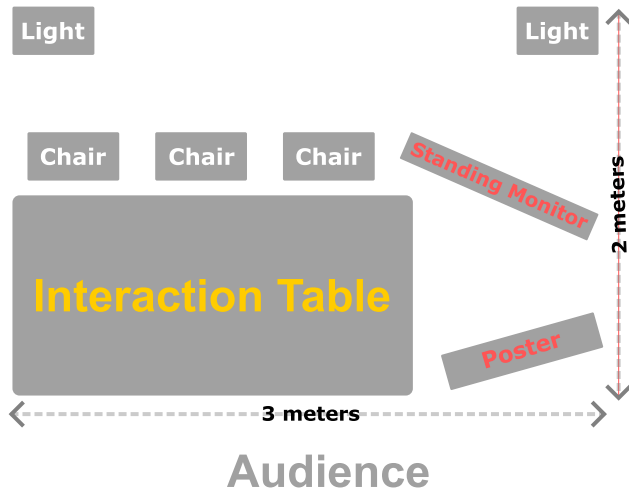


Figure 6: Floor plan of the demonstration setup. Interaction mainly happens at the table, while the poster and the monitor serve the purpose of attracting audience.

- Interaction area: Multiple sets of the VibraForge toolkit will be laid out on a table, allowing attendees to physically interact with the components. Each set will include all necessary parts and tools for assembly and application. Each toolkit will be taken care by a student from the team, ensuring that the audience receive proper guidance.

3.2 Interactive Experience

To maximize engagement and demonstrate the toolkit’s capabilities, the interactive experience will contain the following steps:

- (1) Self-Guided Assembly: Attendees will be invited to assemble their own vibrotactile device using the toolkit components provided. In case they want to follow existing designs, we will also provide some design templates such as wristbands, headbands, etc.
- (2) Customization and Application: Once assembled, participants can mount the device on their body at various locations, and move around to test the wearable comfort level.
- (3) Design and Experimentation: Using the GUI editor, participants will design vibration patterns. This interface will allow them to easily select, modify, and apply different vibratory signals and patterns to the assembled device.

3.3 Expected Outcome

We expect that the demonstration will show how the VibraForge toolkit lowers barriers to designing and implementing custom vibrotactile devices, democratizing access to haptic technology. We also expect to gather some insights from external users regarding user interaction flow, system robustness, and potential areas for enhancement. Since the main contribution is a toolkit, we also hope that this hands-on experience will inspire attendees to think about the combination of haptic feedback and their own research, and

further convince them of the toolkit’s utility and encourage its adoption in their own research or product design.

4 CONCLUSION

This demonstration plan presents a comprehensive approach to showcase the VibraForge toolkit, designed to make vibrotactile device creation accessible and adaptable. Through an interactive demonstration space, participants will directly engage with the toolkit’s assembly and customization capabilities, emphasizing its ease of use and versatility. The plan aims to gather user feedback to validate the system’s design and encourage its broader adoption. This demonstration will not only showcase the toolkit’s potential but also facilitate its continuous improvement and integration into diverse applications. Ultimately, this demonstration will serve as a pivotal step in refining the VibraForge toolkit and expanding its reach and impact within the haptic technology community.

ACKNOWLEDGMENTS

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